Data 101 notes

Read the key ideas and exercises about leading with data.

From files to facts

When county leaders hear the word "data," the conversation usually defaults to IT, software updates, or servers. But data is not an IT issue; it is the fundamental currency of modern public administration.

To understand data, we must first unlearn how government traditionally stores information. Historically, counties have operated on "collections of files"—PDFs, scanned forms, locked spreadsheets, and physical paper. A file is essentially a digital piece of paper. It traps information. To answer a simple question using files, a staff member must manually read, summarize, and cross-reference multiple documents.

Data, by contrast, is a collection of individual, trackable units of fact. The location of a pothole, the timestamp of a 911 dispatch, the property value of a specific parcel—these are data points. When information is broken out of static files and treated as data, it can be instantly sorted, filtered, and combined.

Files force a county to look backward and react. Data allows a county to look forward, predict budget shortfalls, anticipate maintenance needs, and intelligently allocate limited resources.

Data is a public trust

The greatest barrier to becoming a data-smart county is not technical; it is territorial. It is common to hear department heads talk about "my data" or "the Sheriff's data" versus "Public Health's data."

To move forward, leadership must establish a fundamental paradigm shift: **Government agencies do not own data. The constituents do.**

County government data is a public asset, generated by and belonging to the taxpayers. The county is merely a **steward** of that asset. A steward has two strict responsibilities:

1. **Extract Maximum Value:** Use the information to improve services, increase efficiency, and save tax dollars.

2. **Protect the Vulnerable:** Fiercely defend the privacy, security, and integrity of the personal, criminal, or sensitive health elements within that information.

When data is understood as a public trust, departmental silos lose their justification. No single agency can hoard information or refuse to share it with other county departments under the guise of "ownership." You are co-stewards of the public's information, and sharing it securely across operational lines is a requirement of public service, not a favor.

How County data works

You do not need to be a software engineer to lead a data-driven county, but you do need a shared vocabulary of how data moves.

• **How Information is Structured:** The goal is to move from *unstructured* information (like a scanned PDF of a permit application) to *structured* data (a digital portal with dropdown menus). Structured data can be counted and mapped; unstructured data must be read by a human.

• **How We Interface with Data:**

• *Transactional Systems:* The software your frontline staff uses to input facts (e.g., dispatch systems, permitting software, point-of-sale systems).

• *Analytical Systems (Dashboards):* Visual tools like charts and maps used by leadership to see the big picture in real-time, pulling facts from the transactional systems.

• **Data Hygiene ("Garbage In, Garbage Out"):** Data hygiene is the discipline of ensuring information is accurate, complete, and standard across the county (e.g., deciding whether everyone uses "Main Street," "Main St.", or "Mn St."). Poor hygiene isn't just an inefficiency—it leads to bad policy decisions and represents a massive legal and privacy liability if sensitive records are mismanaged or duplicated.

Make evidence safe to share

When data initiatives stall, it is rarely a failure of technology. It is a failure of culture. Data shines a bright light on operational realities, and in government, exposing inefficiencies can feel politically dangerous.

To build a data-smart county, leadership must actively manage the behavioral reality of data:

• **Decouple Data from Punishment:** Leaders are often afraid of what a dashboard might reveal to the board or the press. If a department's data reveals they are failing to meet a metric, executive leadership must reward the transparency rather than punish the failure. If data is weaponized, staff will immediately begin hiding, manipulating, or refusing to collect it.

• **The "What's In It For Me?" (WIIFM):** Data makes the painful parts of leadership easier. It is the ultimate tool for defending budget requests, proving regulatory compliance, winning competitive state and federal grants, and neutralizing angry citizens at board meetings with objective, unassailable facts.

• **The "Show Me" Culture:** The shift to data-driven government starts in the conference room. Executives must refuse to make decisions based on anecdotes, "gut feelings," or whoever complains the loudest. When a department head claims they are severely understaffed, the universal leadership response must become: *"Show me the data that leads you to that conclusion."*

Builders, interpreters, and stewards

Managing servers (IT) is fundamentally different from generating insights from information (Data). As you build this capability, you must source the right talent efficiently so you do not permanently inflate your headcount for temporary needs.

• **The Builders (Data Engineers):** These are the plumbers who build the secure pipes to move data from isolated department software into a central hub. This is heavy, highly technical lifting.

• *Sourcing:* Primarily **Contractors or Specialized Vendors** for the initial architectural build. *If leveraging talented existing staff:* Do so, but plan strategically for their retention and eventual replacement. You must mandate that they adhere to industry standards, require rigorous documentation before launch, and ensure what they build aligns with what your long-term IT infrastructure can actually support.

• **The Interpreters (Data Analysts):** These are the translators who turn raw numbers into dashboards, maps, and answers for leadership.

• *Sourcing:* **Internal Staff.** Analysts require deep, ongoing context regarding county operations, goals, and internal politics to know *what* questions to ask.

• **The Enforcers (Data Stewards):** These are the guardians of data hygiene at the point of entry.

• *Sourcing:* **Existing Staff.** This is not a new hire; it is a shift in daily responsibility for current department heads, frontline managers, and clerks to ensure data is entered accurately.

Start even smaller

Transforming a county government's relationship with data does not require a multi-million-dollar technology overhaul. In fact, attempting to "boil the ocean" usually results in slow, expensive failure.

Instead, start small by changing executive micro-habits.

• **Change the Meeting Currency:** For your next agenda, require one piece of structured, operational data to justify the biggest decision on the table.

• **Redesign the Ask:** Mandate that all future requests for new headcount, equipment, or expanded programs be accompanied by a structured data narrative. If there is no data to prove the need, the request is not ready for review.

Finally, **embrace the mess.** The first time you pull your county's data into the light, it will be ugly, incomplete, and full of errors. That is not a sign of failure; it is step one. The organizations that succeed are the ones brave enough to look at the ugly data and begin cleaning it up, one fact at a time.

Bring better evidence to the next meeting

The next significant budget or service decision is an opportunity to change the conversation. Choose one decision on an upcoming agenda. Before discussing a recommendation, ask the responsible manager to bring a short, structured account of the evidence.

1. **What decision are we making?** Name the service problem, proposed action, and who will be affected.
2. **What facts describe the need?** Ask for counts, trends, and the period measured, not only an anecdote.
3. **How reliable are those facts?** Identify missing entries, inconsistent definitions, and assumptions.
4. **What alternatives were considered?** Compare the request with changes to current practices or resources.
5. **What will we measure afterward?** Choose an outcome that will indicate whether the decision worked.

If the evidence includes individual-level or sensitive records, ask for an appropriate aggregate or otherwise authorized view. The goal is a better decision, not unrestricted circulation of sensitive information.

A first-month leadership challenge

A data-smart culture can begin with a change in leadership habits, not a procurement. Use this illustrative first-month plan as a discussion exercise.

When	Leader's action	What to look for
First meeting	Require one operational fact for a major agenda decision.	The fact has a clear definition and time period.
Next discussion	Ask how the fact was collected and where it may be incomplete.	Staff can surface limitations without being blamed.
Next review	Identify who enters the information, who checks quality, and who interprets results.	Responsibilities are understood across teams.
End of month	Revisit one decision using a consistent measure.	The team can explain what changed and what remains uncertain.

Treat an imperfect first report as the starting point for improvement. The immediate goal is not a perfect dashboard. It is a repeatable expectation that important decisions will be supported by evidence.